

Design Specification: TRUST Tech Deliverable

4a. Enable user GitLab & Docker images

Revision History NA

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1. Introduction

1.1 References

- GovTech GCC Security Guidelines
 - AWS Best Practices for Multi-Account Strategy
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2. Application Overview

2.1 Layers or architectural framework

The solution follows a multi-account AWS architecture for hosting CloudOS and related infrastructure components: * AWS Service Account: Hosts CloudOS and GitLab Server for private code repositories. * AWS Workload Account: Hosts AWS Batch and AWS ECR for containerized workload execution.

2.2 Deployment View

- GitLab Server: Deployed in the AWS Service Account, configured for private access.
- AWS ECR: Deployed in the AWS Workload Account, accessible only within AWS GCC accounts.
- AWS Batch: Utilized for running containerized jobs, with required IAM permissions.

2.3 Development View

- GitLab provides private repository hosting for researchers and engineers.
- CI/CD workflows will be managed within GitLab to integrate with AWS Batch.
- AWS ECR will store and manage private container images for workloads.

2.4 Architectural Mechanisms

- Secure networking between AWS accounts using VPC peering or AWS PrivateLink.
 - IAM roles and policies restricting access to only necessary resources.
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3. Architecturally significant requirements

- Deployment in a non-internet environment within Singapore GCC.

- Secure and private access for all services.
 - Multi-account strategy for isolation and compliance.
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4. Security

4.1 System Security

- Network access controls enforced via Security Groups and VPC.
- AWS KMS for encrypting sensitive data and images.

4.2 Data Security

- No external data access permitted; all data resides within AWS GCC.
- Private container images stored in AWS ECR with strict access policies.

4.3 Application Security

- GitLab access restricted to authorized users only.
 - IAM roles configured for least privilege access.
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5. Common Modules

- NA
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6. Functional Modules

- NA
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7. Message / Service Interface

- NA
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8. Database Design

- NA
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Appendix A

- NA